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Ever wonder how top tech companies like Netflix, Uber, and Airbnb measure impact when standard A/B testing is impossible?

Whether it's a nationwide rollout, a marketing campaign, or macroeconomic shifts, randomizing users isn't always an option. Enter Quasi-Experiments!

Here are 4 of the most powerful causal inference methods used by world-class data teams:

Difference-in-Differences (DiD)

DiD compares the changes in outcomes over time between a treatment group and a control group. By observing both groups before and after an event, it filters out natural, pre-existing trends to isolate the true impact of an intervention.

DoorDash uses DiD combined with causal inference to calculate the true impact of macroeconomic factors (like daylight saving time or tax refunds) on delivery demand, separating the actual signal from natural noise.

<https://lnkd.in/dZSHRVbj>

Synthetic Control Method (SCM)

What if you launch a feature in just one city and there is no identical city to compare it to? SCM takes a weighted combination of several untreated cities to build an artificial, "synthetic" control group that perfectly matches your target market's historical trends.

Netflix relies on synthetic controls to evaluate the impact of highly localized rollouts and out-of-home advertising campaigns where traditional user-level A/B testing falls short.

<https://lnkd.in/dz5sKqgA>

Propensity Score Matching (PSM)

When users self-select into a feature, you face massive selection bias. PSM solves this by finding a statistical "twin" in the control group for every treated user. It scores them based on their characteristics to ensure an apples-to-apples comparison.

Airbnb uses PSM heavily to construct counterfactuals when analyzing how specific guest or host actions impact the future lifetime value on their platform.

<https://lnkd.in/dz-P7vQj>

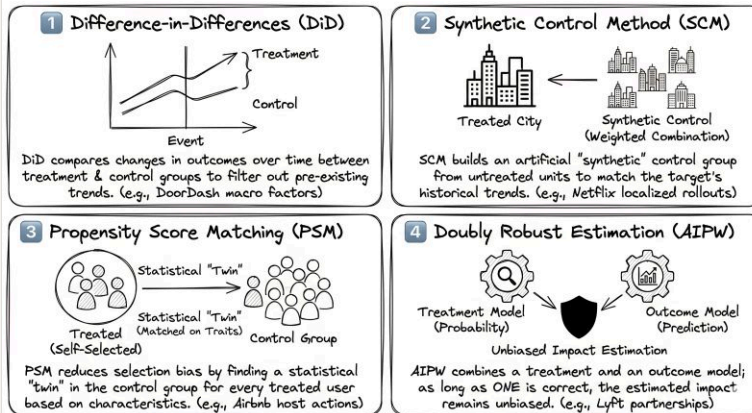
Doubly Robust Estimation (AIPW)

Augmented Inverse Propensity Weighting (AIPW) combines two models: one that predicts the probability of treatment, and another that predicts the outcome. It's called "doubly robust" because as long as at least ONE of these models is correct, your estimated impact remains unbiased!

Lyft leverages doubly robust models in their experimentation platform to accurately measure the impact of unrandomized business partnerships and long-term market policies.

<https://lnkd.in/dXDJDRJV>

I'm noticing that causal inference is becoming essential for anyone working with experiments. Which methods do you use the most?



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- Michael Kaminsky** · 2nd 1mo ...
Recast Co-Founder | Writes about marketing science, incrementality, and ...
Great overview! For both PSM and Synthetic Controls (and presumably AIPW) I think it's important to have fully algorithmic methods that are applied in order to reduce the chance of analyst bias. They're great tools but they also open up a lot of degrees of freedom for the analyst to shape the results if you're not careful!
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- Yuriy Balandin** **Author** 1mo ...
Senior Data Scientist | A/B Testing & Causal Inference | Machine Lea...
[Michael Kaminsky](#) absolutely, it's crucial to check assumptions for each of the methods and ideally conduct simulations on the historic data to check false positives and power
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- Krishnaja Koonapra** · 3rd+ (edited) 1mo ...
Data Scientist, Product Analytics @ Twinkl | Data Analyst | Deliver...
Thanks for this [Yuriy Balandin](#) !!sorted my reading for today
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